

Binomial Coefficients

COMP2711 Discrete Mathematical Tools

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Example: Expanding a cube

What is the coefficient of xy^2 in $(x + y)^3$?

$$(x + y)^3 = \underbrace{(x + y)}_{\text{factor 1}} \underbrace{(x + y)}_{\text{factor 2}} \underbrace{(x + y)}_{\text{factor 3}}.$$

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$$\begin{aligned} &= (x^2 + xy + yx + y^2)(x + y) \\ &= x^3 + x^2y + x^2y + \mathbf{xy^2} + x^2y + \mathbf{xy^2} + \mathbf{xy^2} + y^3 \\ &= x^3 + 3x^2y + \mathbf{3xy^2} + y^3. \end{aligned}$$

The coefficient of xy^2 is **3**.

Example: Counting the coefficient

Can we find the same coefficient of xy^2 **without expanding every term**?

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x	y	y	→	xy^2
y	x	y	→	xy^2
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Three choices produce xy^2 , so its coefficient is **3**.

From coefficients to counting identities

Choose terms from factors

Find coefficients without expanding everything.

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Count the same objects in two ways

Explain identities using subsets and teams.

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Substitute constants in the binomial theorem

Obtain ordinary, alternating, and weighted sums.

SECTION 1

The binomial theorem

The binomial theorem

MUST UNDERSTAND

The same factor-choice argument works for any integer $n \geq 0$:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

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The product

k factors supply y .

The other $n - k$ supply x .

The coefficient

Choose those k factors:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

The binomial theorem

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The same factor-choice argument works for any integer $n \geq 0$:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

For constants a, b , replace x, y by ax, by :

$$(ax + by)^n = \sum_{k=0}^n \binom{n}{k} (ax)^{n-k} (by)^k = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k x^{n-k} y^k.$$

Example 1: A full expansion

Use the theorem to expand $(1 + x)^4$.

k factors supply x	Number of choices	Term
0	$\binom{4}{0} = 1$	1
1	$\binom{4}{1} = 4$	$4x$

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2	$\binom{4}{2} = 6$	$6x^2$
3	$\binom{4}{3} = 4$	$4x^3$
4	$\binom{4}{4} = 1$	x^4

Example 1: A full expansion

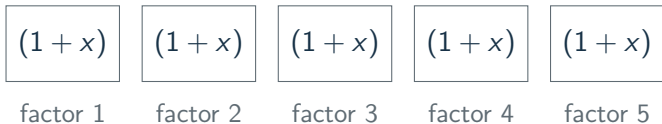
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$$(1 + x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4.$$

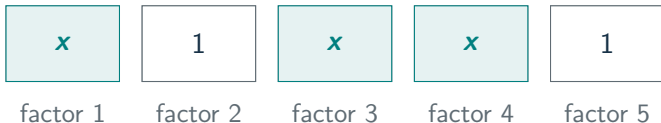
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Without expanding everything, find the coefficient of x^3 in $(1 + x)^5$.



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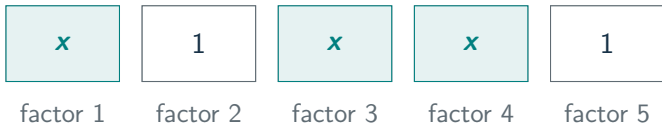
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Choose **three of the five factors** to supply x . The others supply 1 .

Example 2: One coefficient

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$$\text{Coefficient} = \binom{5}{3} = \frac{5 \cdot 4 \cdot 3}{3 \cdot 2 \cdot 1} = \mathbf{10}.$$

Example 3: Constants and signs

Now include constants and a minus sign: find the coefficient of x^5 in $(2x - 1)^8$.

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Now include constants and a minus sign: find the coefficient of x^5 in $(2x - 1)^8$.

Let k factors supply -1 ; the other $8 - k$ supply $2x$.

$$\underbrace{\binom{8}{k}}_{\text{choose factors}} \underbrace{(2x)^{8-k}}_{8-k \text{ factors}} \underbrace{(-1)^k}_{k \text{ factors}}.$$

Example 3: Matching the exponent

Find the coefficient of x^5 in $(2x - 1)^8$.

$$\binom{8}{k} (2x)^{8-k} (-1)^k = \binom{8}{k} 2^{8-k} (-1)^k x^{8-k}.$$

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Match the exponent: $8 - k = 5$, so $k = 3$.

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Match the exponent: $8 - k = 5$, so $k = 3$.

$$\begin{aligned}\text{Coefficient} &= \binom{8}{3} 2^5 (-1)^3 \\ &= 56 \cdot 32 \cdot (-1) = -1792.\end{aligned}$$

Coefficient extraction

MUST UNDERSTAND

1. Write the general term.

$$(ax + by)^n : \quad \binom{n}{k} (ax)^{n-k} (by)^k.$$

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Coefficient extraction

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1. Write the general term.
2. Match the target exponent.
3. Evaluate the coefficient, including constants and signs.

$$(ax + by)^n : \quad \binom{n}{k} (ax)^{n-k} (by)^k.$$

The index must be an integer with $0 \leq k \leq n$.

No valid index $\implies$ the requested coefficient is 0.

Example 4: A constant term

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$$\binom{9}{k} (x^2)^{9-k} (x^{-1})^k = \binom{9}{k} x^{18-3k}.$$

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A constant has exponent 0:

$$18 - 3k = 0 \quad \implies \quad k = 6.$$

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$$\text{Constant term} = \binom{9}{6} = 84.$$

Practice: An absent term

In the same expansion, $(x^2 + x^{-1})^9$, what is the coefficient of x^4 ?

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Practice: An absent term

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$$\text{General term} = \binom{9}{k} x^{18-3k}.$$

$$18 - 3k = 4 \implies k = \frac{14}{3}.$$

This is not an integer: no choice of factors produces x^4 .

$$\text{Coefficient} = \mathbf{0}.$$

SECTION 2

Counting identities

Combinatorial proofs

MUST UNDERSTAND

Counting gave us the binomial coefficients. It can also explain identities between them.

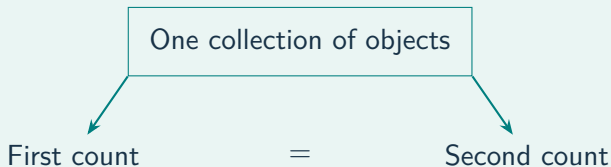
A **combinatorial proof** establishes an identity by counting **the same finite collection in two ways**.

Combinatorial proofs

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Counting gave us the binomial coefficients. It can also explain identities between them.

A **combinatorial proof** establishes an identity by counting **the same finite collection in two ways**.



Both counts must include every object **exactly once**.

Three ways to split a selection

Choose k objects from n . The total is $\binom{n}{k}$.

Identity	Formula
Pascal	$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$
Hockey-stick	$\binom{n}{k} = \sum_{j=k}^n \binom{j-1}{k-1}$
Vandermonde	$\binom{n}{k} = \sum_{i=0}^k \binom{a}{i} \binom{b}{k-i} \quad (a+b=n)$

Here $1 \leq k < n$; a, b are nonnegative integers with $a + b = n$.

Running example: Choose four from six

Choose a four-person team from $\{A, B, C, D, E, F\}$.



How many different teams can we form?

Equivalent View 1: Include or exclude A

Direct count: $\binom{6}{4} = 15$. Split the same teams by whether A is selected.

A is selected

Choose three from B, C, D, E, F .

$$\binom{5}{3} = 10.$$

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Choose four from B, C, D, E, F .

$$\binom{5}{4} = 5.$$

Every team belongs to exactly one case.

$$\binom{6}{4} = \binom{5}{3} + \binom{5}{4} = 10 + 5 = 15.$$

Equivalent View 1: Pascal's identity

Generalize the team example: choose k from n , where $n \geq 2$ and $1 \leq k < n$.

$$\underbrace{\binom{n}{k}}_{\text{all } k\text{-person teams}} = \underbrace{\binom{n-1}{k-1}}_{\text{teams including } A} + \underbrace{\binom{n-1}{k}}_{\text{teams excluding } A} .$$

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Include a fixed person A

Choose $k - 1$ more from the other $n - 1$ people.

Exclude A

Choose all k from the other $n - 1$ people.

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Choose all k from the other $n - 1$ people.

Boundary values: $\binom{n}{0} = \binom{n}{n} = 1.$

Pascal's triangle

row 0	1					
row 1	1	1				
row 2	1	2	1			
row 3	1	3	3	1		
row 4	1	4	6	4	1	
row 5	1	5	10	10	5	1

Pascal's identity

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Row n lists $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$: the coefficients of $(x + y)^n$.

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row 5	1	5	10	10	5	1	
row 6	1	6	15	20	15	6	1

10 + 5 = 15

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$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Row n lists $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$: the coefficients of $(x + y)^n$.

Equivalent View 2: The largest label

Keep the same four-person teams; change the cases. Label $A, \dots, F$ as $1, \dots, 6$.

Split the teams by their **largest selected label**.

$\{1, 2, 3, 4\}$

largest: 4

$\{1, 2, 4, 5\}$

largest: 5

$\{2, 3, 5, 6\}$

largest: 6

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$\{2, 3, 5, 6\}$

largest: 6

Every selection has exactly one largest number.

Its possible values are 4, 5, 6.

Equivalent View 2: Count each case

Fix the largest label, then choose the other three from smaller labels.

Largest label	Choose the other three from	Count
4	$\{1, 2, 3\}$	$\binom{3}{3} = 1$

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Every subset belongs to exactly one case.

$$\binom{6}{4} = \binom{3}{3} + \binom{4}{3} + \binom{5}{3} = 1 + 4 + 10 = 15.$$

Equivalent View 2: Hockey-stick identity

Generalize the largest-label count: choose k from $\{1, \dots, n\}$, where $1 \leq k \leq n$.

$$\boxed{\binom{n}{k} = \sum_{j=k}^n \binom{j-1}{k-1}}.$$

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Case: the largest number is j .

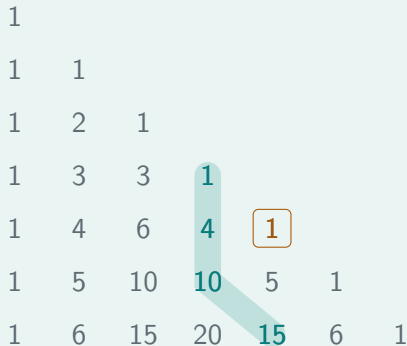
Fix j , then choose the other $k-1$ numbers from $\{1, \dots, j-1\}$.

The cases $j = k, \dots, n$ cover every subset exactly once.



Hockey-stick and Pascal's identities

The same sum $1 + 4 + 10 = 15$ also follows by repeating Pascal's rule.



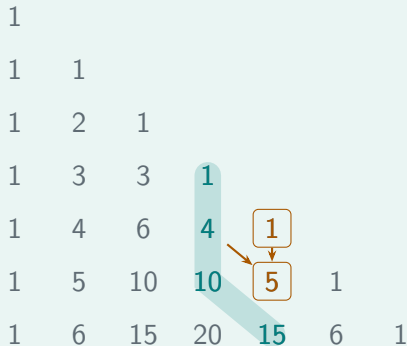
1						
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Start with $\binom{3}{3} = \binom{4}{4} = 1$.

Add the next highlighted entry:

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1						
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1	5	10	10	5	1	
1	6	15	20	15	6	1

Start with $\binom{3}{3} = \binom{4}{4} = 1$.

Add the next highlighted entry:

$$1 + 4 = 5 = \binom{5}{4}$$

$$1 + 4 + 10 = 15 = \binom{6}{4}.$$

The hockey-stick sum comes from applying Pascal's rule repeatedly.

Equivalent View 3: Two groups

Keep the same four-person teams; now split the six people into two groups of three.



Group 1: three



Group 2: three

Split the teams by how many members come from Group 1.

Equivalent View 3: Count each case

Direct count: $\binom{6}{4} = 15$. Count by the number chosen from Group 1.

From Group 1	From Group 2	Choices	Count
1	3	$\binom{3}{1}\binom{3}{3}$	3

Equivalent View 3: Count each case

Direct count: $\binom{6}{4} = 15$. Count by the number chosen from Group 1.

From Group 1	From Group 2	Choices	Count
1	3	$\binom{3}{1} \binom{3}{3}$	3
2	2	$\binom{3}{2} \binom{3}{2}$	9
3	1	$\binom{3}{3} \binom{3}{1}$	3

Choosing 0 or 4 from Group 1 is impossible.

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Multiply the choices within each case, then add the case counts.

$$\binom{6}{4} = \binom{3}{1}\binom{3}{3} + \binom{3}{2}\binom{3}{2} + \binom{3}{3}\binom{3}{1} = 3 + 9 + 3 = 15.$$

Equivalent View 3: Vandermonde's identity

Generalize to two disjoint groups of sizes $a, b \geq 0$. Choose k people from $n = a + b$, where $0 \leq k \leq n$.

$$\binom{n}{k} = \binom{a+b}{k} = \sum_{i=0}^k \binom{a}{i} \binom{b}{k-i}.$$

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Generalize to two disjoint groups of sizes $a, b \geq 0$. Choose k people from $n = a + b$, where $0 \leq k \leq n$.

$$\binom{n}{k} = \binom{a+b}{k} = \sum_{i=0}^k \binom{a}{i} \binom{b}{k-i}.$$

Case: i people come from Group 1.

Choose i from the first group and $k - i$ from the second.

An impossible choice has count 0: $\binom{u}{v} = 0$ if $v < 0$ or $v > u$.

Pascal as a special case of Vandermonde

Return to View 1: put A alone in Group 1 and the other $n - 1$ people in Group 2.

Choose a k -person team, where $1 \leq k < n$.

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$$\binom{n}{k} = \underbrace{\binom{1}{0} \binom{n-1}{k}}_{\text{exclude } A} + \underbrace{\binom{1}{1} \binom{n-1}{k-1}}_{\text{include } A}.$$

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Choose a k -person team, where $1 \leq k < n$.

$$\binom{n}{k} = \underbrace{\binom{1}{0} \binom{n-1}{k}}_{\text{exclude } A} + \underbrace{\binom{1}{1} \binom{n-1}{k-1}}_{\text{include } A}.$$

Only $i = 0$ and $i = 1$ are possible. Since $\binom{1}{0} = \binom{1}{1} = 1$,

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

Application: A sum of squares

Another special case: two equal groups of size n , with n people selected in total.

$$\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k} \binom{n}{n-k}.$$

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By the complement identity, $\binom{n}{n-k} = \binom{n}{k}$.

$$\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k}^2.$$

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For $n = 3$:

$$\binom{6}{3} = 1^2 + 3^2 + 3^2 + 1^2 = 20.$$

SECTION 3

Corollaries of the Binomial Theorem

Substituting constants

We counted selections in different ways. Now use algebra: for integers $n \geq 0$,

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

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We counted selections in different ways. Now use algebra: for integers $n \geq 0$,

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What happens when we choose constants for x and y ?

x	y	Resulting identity			
1	1	$(1 + 1)^n$	$=$	2^n	$= \sum_{k=0}^n \binom{n}{k}$
1	-1	$(1 - 1)^n$	$=$	0	$= \sum_{k=0}^n (-1)^k \binom{n}{k} \quad (n \geq 1)$
1	2	$(1 + 2)^n$	$=$	3^n	$= \sum_{k=0}^n 2^k \binom{n}{k}$

Example: All subsets

Let $S = \{A, B, C\}$.

How many subsets does S have, including the empty set?

Solution: All subsets

For the ordinary sum, count the subsets of $S = \{A, B, C\}$ by size:

Size	Subsets	Count
0	$\emptyset$	1
1	$\{A\}, \{B\}, \{C\}$	3
2	$\{A, B\}, \{A, C\}, \{B, C\}$	3
3	$\{A, B, C\}$	1

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3	$\{A, B, C\}$	1

For $\{A, B, C\}$, add the counts for all subset sizes:

$$\underbrace{1 + 3 + 3 + 1}_{\text{group by size}} = 8 = \underbrace{2 \cdot 2 \cdot 2}_{\text{include or exclude each element}}.$$

Sum of binomial coefficients

Generalize from three elements to n elements, where $n \geq 0$:

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

Both sides count all subsets of an n -element set.

- Left: group subsets by their size k .
- Right: include or exclude each of the n elements.

Sum of binomial coefficients: Algebraic proof

Recover the same subset count from the binomial theorem, for integers $n \geq 0$:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

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Only the binomial coefficients remain:

$$\boxed{2^n = \sum_{k=0}^n \binom{n}{k}}.$$

Example: Even and odd subsets

Now compare subset sizes: let S be a nonempty finite set.

Does S have more even-size subsets or more odd-size subsets?

Solution: Even and odd subsets

Let's construct a **bijection** between all even-size subsets and all odd-size subsets.

Let S be a nonempty finite set. Fix an element $A \in S$.

Example: $S = \{A, B, C\}$

Even size		Odd size
$\emptyset$	$\longrightarrow$	$\{A\}$
$\{A, B\}$	$\longrightarrow$	$\{B\}$
$\{A, C\}$	$\longrightarrow$	$\{C\}$
$\{B, C\}$	$\longrightarrow$	$\{A, B, C\}$

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Define the map

For every $T \subseteq S$ with $|T|$ even,

$$f(T) = \begin{cases} T \setminus \{A\}, & A \in T, \\ T \cup \{A\}, & A \notin T. \end{cases}$$

Add or remove A ; the result has odd size.

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Add or remove A ; the result has odd size.

Exercise: Show that f is a bijection.

Alternating sum

The bijection makes the two counts equal for $n \geq 1$ (sizes up to n):

$$\underbrace{\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots}_{\text{even sizes}} = \underbrace{\binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots}_{\text{odd sizes}} = 2^{n-1}.$$

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$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \cdots = 0.$$

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$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \cdots = 0.$$

The sign is $+1$ for even k and -1 for odd k : exactly $(-1)^k$.

$$\sum_{k=0}^n (-1)^k \binom{n}{k} = 0.$$

Alternating sum: Algebraic proof

Recover the same alternating signs by setting $x = 1$ and $y = -1$:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

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$$\boxed{0 = \sum_{k=0}^n (-1)^k \binom{n}{k} .}$$

Example: Two labels or no label

Each of three students gets a red label, a blue label, or no label.



How many different label assignments are possible?

Solution: Two labels or no label

Each of the three students has three choices:

$$3^3 = 27 \text{ assignments.}$$

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Count by recipients: $\binom{3}{k}$ ways to choose them, then 2^k colorings.

$$\binom{3}{0}2^0 + \binom{3}{1}2^1 + \binom{3}{2}2^2 + \binom{3}{3}2^3 = 1 + 6 + 12 + 8 = 27.$$

A weighted sum

The same labeling argument for n students gives, for $n \geq 0$,

$$\sum_{k=0}^n \binom{n}{k} 2^k = 3^n.$$

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Recipients first

Choose k students: $\binom{n}{k}$.

Choose their colors: 2^k .

Algebra: substitute $x = 1$, $y = 2$ into $(x + y)^n$.

One student at a time

Each has three states:

red, blue, or no label.

Practice: Three colors or no label

Now allow three colors: each of n students gets one color or no label, where $n \geq 0$.

How many different label assignments are possible?

Solution: Three colors or no label

Each student has four choices, so there are 4^n assignments.

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Or choose k recipients, then one of three colors for each:

$$\sum_{k=0}^n \underbrace{\binom{n}{k}}_{\text{choose recipients}} \underbrace{3^k}_{\text{choose their colors}} = 4^n.$$

SECTION 4

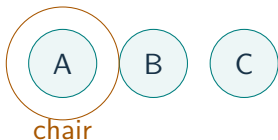
Counting proofs

Challenge: A committee and its chair

There are n people, where $n \geq 1$.

How many ways can we choose a nonempty committee and one of its members as chair?

The committee can have any size from 1 to n .



Same members, different chair
 $\Rightarrow$ different outcome.

Solution: Two counting orders

Again choose a subset, then distinguish one member as chair.

Two ways to count:

1. Choose the committee, then its chair.
2. Choose the chair, then the other committee members.

Solution: Two counting orders

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Two ways to count:

1. Choose the committee, then its chair.
2. Choose the chair, then the other committee members.

Both count the same committees with a chair, so their answers must be equal.

Challenge: Committee first

Method 1: choose k members, then choose their chair.

$$\underbrace{\binom{n}{k}}_{\text{choose members}} \underbrace{k}_{\text{choose their chair}} .$$

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For $n = 3$:

Committee size	Member choices	Chair choices	Outcomes
1	3	1	3
2	3	2	6
3	1	3	3

Challenge: Committee first

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Committee size	Member choices	Chair choices	Outcomes
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2	3	2	6
3	1	3	3

Add all sizes $k = 1, \dots, n$: $\sum_{k=1}^n k \binom{n}{k}$. For $n = 3$: 12.

Challenge: Chair first

Method 2: reverse the order. Choose the chair first: n choices.



Each of the other $n - 1$ people is independently included or excluded.

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$$\text{Count} = n \cdot 2^{n-1}; \quad n = 3 : \quad 3 \cdot 4 = 12.$$

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$$\text{Count} = n \cdot 2^{n-1}; \quad n = 3 : \quad 3 \cdot 4 = 12.$$

Both methods count the same chaired committees:

$$\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}.$$

Optional: Small subsets

Let $S = \{1, 2, 3, 4, 5\}$.

How many subsets of S have at most two elements?

Solution: Small subsets

Use pairing again: pair each subset of $S = \{1, 2, 3, 4, 5\}$ with its complement.

Subset size		Complement size
0	$\longleftrightarrow$	5
1	$\longleftrightarrow$	4
2	$\longleftrightarrow$	3

Example pair:

$$\{1, 4\} \longleftrightarrow \{2, 3, 5\}.$$

Exactly one subset in each pair has size at most 2.

Solution: Small subsets

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Subset size		Complement size	Example pair:
0	$\longleftrightarrow$	5	$\{1, 4\} \longleftrightarrow \{2, 3, 5\}.$
1	$\longleftrightarrow$	4	Exactly one subset in each pair has size at most 2.
2	$\longleftrightarrow$	3	

$$\text{Small subsets} = \frac{2^5}{2} = 16 = \binom{5}{0} + \binom{5}{1} + \binom{5}{2}.$$

Optional: Complement pairing

Generalize from five elements to any odd size $|S| = 2n - 1$, where $n \geq 1$.

Pair $A \subseteq S$ with $S \setminus A$.

$$|A| + |S \setminus A| = 2n - 1.$$

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The sizes cannot tie: exactly one is at most $n - 1$.

So half of all 2^{2n-1} subsets have size at most $n - 1$.

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So half of all 2^{2n-1} subsets have size at most $n - 1$.

$$\sum_{k=0}^{n-1} \binom{2n-1}{k} = 2^{2n-2}.$$

Odd size matters. With an even-size ground set, complements can have equal sizes.

Takeaways

- **Find a coefficient:** write the general term and match its exponent.

Next: inclusion-exclusion.

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- **Find a coefficient:** write the general term and match its exponent.
- **Prove an identity:** name one collection and count it in two ways.
- **Get corollaries:** substitute constants into the binomial theorem.

Next: inclusion-exclusion.

Formula summary

Binomial theorem	$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$
Pascal's identity	$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k} \quad (1 \leq k < n)$
Hockey-stick identity	$\binom{n}{k} = \sum_{j=k}^n \binom{j-1}{k-1} \quad (1 \leq k \leq n)$
Vandermonde's identity	$\binom{a+b}{k} = \sum_{i=0}^k \binom{a}{i} \binom{b}{k-i}$
Sum of squares	$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$
All subsets	$\sum_{k=0}^n \binom{n}{k} = 2^n$
Even-size vs. odd-size subsets	$\sum_{k=0}^n (-1)^k \binom{n}{k} = 0 \quad (n \geq 1)$
Two labels or no label	$\sum_{k=0}^n 2^k \binom{n}{k} = 3^n$
Committee and chair	$\sum_{k=1}^n k \binom{n}{k} = n 2^{n-1} \quad (n \geq 1)$
Complement pairing (optional)	$\sum_{k=0}^{n-1} \binom{2n-1}{k} = 2^{2n-2} \quad (n \geq 1)$

Parameters are nonnegative integers; $\binom{a}{b} = 0$ when $b < 0$ or $b > a$.